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- The diagram illustrates the structure of a mitochondrion. The outer mitochondrial membrane is smooth and separates the cytoplasm from the intermembrane space. The inner mitochondrial membrane is folded into cristae and separates the intermembrane space from the mitochondrial matrix. Various proteins are embedded in both membranes, labeled A, B, C, and D. The intermembrane space is the region between the two membranes, and the mitochondrial matrix is the fluid-filled space inside the inner membrane.

On Fig. 6.1, use arrows to show the route taken by oxygen from the cytoplasm into the mitochondrial matrix. [1]

- [7]



- (c) The matrix of a mitochondrion contains small circular DNA. In the cells of humans, the mitochondria have circular DNA that contains 37 genes.

Suggest functions for these mitochondrial genes.

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..... [3]

[Total: 11]